Campus Food Ordering System

A Multi-Tenant Web & Cross-Platform Mobile Application

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Introduction

Campus canteens and food courts face severe operational bottlenecks during peak lunch hours, creating an urgent need for an automated, real-time digital solution.

- Long queues (40–60 students per stall) during short recess breaks
- Manual paper-based slips prone to order mix-ups and billing errors
- Zero real-time visibility into kitchen queue and preparation times
- Cash transactions leading to change shortages and delayed handoffs

This project presents a **full-stack multi-tenant platform** providing contactless ordering, real-time kitchen orchestration, and PIN-verified delivery across multiple campus vendors from a single cloud instance.

Objectives

- **Reduce Queue Wait Time:** Cut peak waiting time by 60% via digital pre-ordering.
- **Multi-Tenant Architecture:** Power multiple campus food stalls from a single shared backend.
- **Role-Specific Portals:** Deliver tailored interfaces for Customer, Kitchen, Delivery, and Admin.
- **Live Order Tracking:** Enable real-time order lifecycle events using Socket.io WebSockets.
- **Cross-Platform Access:** Provide responsive web app (React) and mobile app (React Native).
- **Secure Operations:** Role-based access control with JWT authentication and secure data isolation.

Methodology

Developed using **Agile Scrum** over six 2-week iterations with end-of-sprint stakeholder reviews:

- **Sprint 1–2:** JWT auth, role guards, menu CRUD & Cloudinary CDN image uploads
- **Sprint 3–4:** Cart checkout, order state machine & live Socket.io kitchen dashboard
- **Sprint 5–6:** Delivery agent panel with PIN verification & React Native mobile app QA

Frontend: React.js | Redux Toolkit | React Router v6 | Axios
Backend: Node.js | Express.js | MongoDB Atlas | JWT Auth
Services: React Native | Socket.io | Cloudinary | REST API

System Architecture & Data Flow

A **RESTful 3-tier architecture** with bi-directional WebSocket channels: React/Mobile client → Express API gateway → MongoDB persistence. Status updates broadcast instantly without polling.

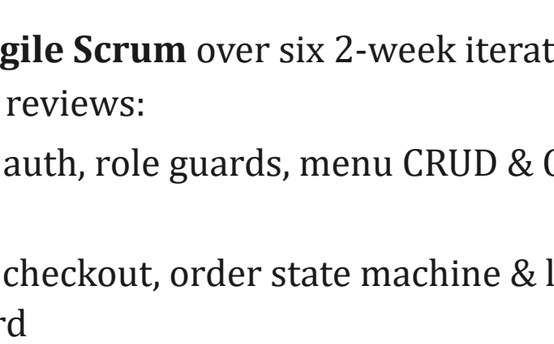
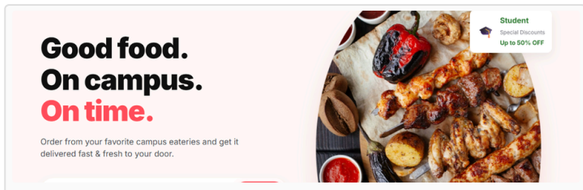


Figure 1: Level-1 Data Flow Diagram — Campus Food Ordering System

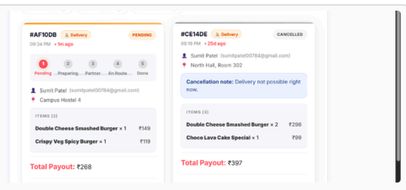
Results & Discussion

60%	<108ms	200+
Queue Time Reduction	Avg. API Latency	Concurrent Users

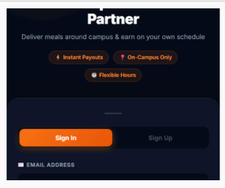
- **Zero cart conflicts:** Multi-tenant isolation verified during high-load concurrent stress tests.
- **100% Delivery Accuracy:** 4-digit PIN handoff eliminated order delivery misplacements.
- **Real-time Synchronization:** Socket.io state propagation achieved within 85ms across devices.



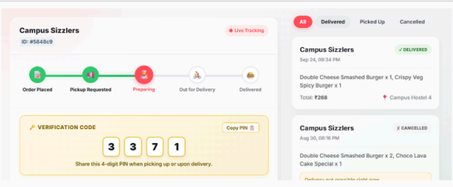
Customer Menu & Cart



Kitchen Live Dashboard



Delivery Agent Panel



Real-Time Tracking View

Figure 2: Production UI — Customer, Kitchen, Delivery & Live Tracking

Conclusion & Future Work

The platform successfully replaces chaotic counter crowding with structured digital order workflows. Scalable cloud architecture allows seamless onboarding of new food vendors with zero infrastructure changes.

- **Key Impact:** Drastic reduction in wait times, zero billing errors, automated kitchen queues.
- **Applications:** College food courts, corporate dining complexes, hospital cafeterias.
- **Limitations:** Requires active on-campus network; currently operates in English language.

Future Enhancements:

- Integration with UPI / Razorpay payment gateways for instant contactless checkout
- AI-powered demand forecasting to optimize stall inventory and minimize food waste
- NFC / Student ID card tap-and-pay support with automated meal voucher plans

References

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